Difference Between Shader Model 2.0 And 3.0

The new nVidia 6X series supports Shader model 3.0. What is Shader model 3.0? It is an improvisation over the old Shader Model 2.0 available in the last generation of cards which are the 5X series for nVidia and 95X series for ATi. In fact, ATi still haven't moved to Shader Model 3.0 in their current crop of cards, which is the X8X series.

Technically, Shader model 3.0 offers more in every department over Shader model 2.0.

Given below is a brief table of the improvements of SM3.0 over SM2.0.

We can see that there is hardly any image improvement from SM 2.0 to SM 3.0. For now, it seems that SM 3.0 will probably provide more of a performance gain in games than actual image quality. The variation in image quality can only be measured with the emergence of new games that will implement Shader model 3.0 instructions from scratch.

Pixel Shader Feature	Shader 2.0	Shader 3.0	Description
Shader length	96	65535+	Improvement in image quality
Dynamic branching	No	Yes	Improvement in performance
Shader anti-aliasing	Not supported	Built-in derivative instructions(Anti-aliasing support in game)	Improvement in image quality
Back-face register	No	Yes	Improvement in performance
Interpolated color format	8-bit integer minimum	32-bit floating point minimum	Improved colors leading to improvement in image quality
Multiple render targets	Optional	4 required	Improved visual quality without loss in performance
Fog and specular	8-bit fixed function minimum	Custom fp16-fp32 Shader program	More precise calculations leading to fog and specular lighting being directly controlled by the programmer
Texture coordinate count	8	10	Improvement in visual quality

Vertex Shader feature	Shader 2.0	Shader 3.0	Description
Shader length	256 Instructions	Shader 3.0	Improved visual quality
Dynamic branching	No	65535 instructions	Improved performance
Vertex texture	No	Yes	Improved visual quality
Instancing support	No	Any number of lookups from up to 4 textures Required	Improved performance

which, if you look from above, partitions the motherboard into two perfect halves!

Their Performance

The performance of the high-end cards was the highlight of the comparison. The Tier 1 tests were child's play for these cards. In the *Far Cry* test, the XFX 6800GT trounced the other cards with sheer authority.

It posted a record 138.57 fps at 1,024 x 768 with the game settings set to Medium. *Doom 3* and *Half-Life 2* fared similarly on this card, and we wondered if the manufacturer had hidden a steroid shot somewhere in the card's heat sink! The X800 PRO is relatively underpowered because of its 12-pixel pipeline architecture compared to the 16-pixel pipeline architecture of the 6800GT.

However, that does not mean that the card in any way lacks spunk. In 3DMark, the card posted the highest scores amongst all the other cards, even ahead of the X800XT PE.

Moving on from the Tier 1 tests, we had four distinct winners. These were the Gainward 6800 GT 256 MB, the Gainward 6800 256 MB, the XFX 6800GT 256 MB and the PowerColor X800 PRO.

In our Tier 2 tests, the in-game settings were turned to High as usual. Before we get into the results, let's talk a little about the textures in *Doom 3*. Any gamer who has played *Doom 3* will agree that it has fantastic lighting and that the textures are excellent. In *Doom 3*, at low quality, everything is toned down—lighting and textures—and all the maps are compressed, while all the textures are fixed to 512 x 512 with specular maps fixed to 64 x 64. Anyone with a decent 64 MB video card can play in the low quality mode with relatively good frame-rates.

But as we move on to the High settings, the specular and diffuse maps are compressed using DXTC 1, 3, 5, while the normal maps are uncompressed. In addition, the texture sizes are not fixed.

This puts a major load on the GPU, and the DDR3 RAM on the high-end cards helps in this area since the faster speed lets it load the textures faster.

In the Ultra quality mode, everything is uncompressed, and here the texture data can go above 500 MB, making it suitable for video cards of the future which will feature 512 MB of video RAM. We did try the Ultra High quality on the XFX 6800GT and the Gainward 6800GT.

What we saw was utterly unbelievable! No game has ever looked so good. Everything seemed to jump out of the screen, and given the sound effects, it was truly a visual and aural treat to experience.

Half-Life 2 is a relatively less hungry for resources, and so even when you move on to high quality, the frame-rates dip, but not massively. The game is more about the physics and the engine rather than heavy textures. One place the game stands out is the water. It is amazing and this is where the pixel shaders of the card have to work. The shader-intensive water was quite a match for even the best video card we had.

In *Far Cry* though, effects abound. You name it and, the game has it, in order to look good.

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